



HYPERIONSOFT

AI & Data Services

TECHNICAL CAPABILITIES BRIEF · DRUG DISCOVERY AI

AI-Powered Drug Discovery Services

ADMET Prediction · Drug Repurposing · De Novo Design

Architecture references, model benchmarks, case studies, integration requirements, and engagement models for pharma and biotech organizations.

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TABLE OF CONTENTS

What's Inside

01	Executive Summary	3
02	Technical Architecture & Model Benchmarks	4
03	Case Studies with Quantified Outcomes	6
04	Data Requirements & Integration Checklist	8
05	Engagement Models & Timeline Expectations	10
06	About Hyperionsoft & Academic Affiliation	12



SECTION 01

Executive Summary

The pharmaceutical industry faces a persistent productivity paradox: despite record R&D investment, the average cost to bring a new drug to market has exceeded \$2.6 billion, with attrition rates above 90% in clinical development. The majority of failures trace to issues that are, in principle, predictable — poor pharmacokinetic profiles, off-target toxicity, and an incomplete search of therapeutic chemical space.

Hyperionsoft applies enterprise-grade AI and data infrastructure to three high-impact intervention points in the drug discovery pipeline: ADMET property prediction, computational drug repurposing, and generative de novo molecular design. Our approach combines state-of-the-art deep learning architectures with rigorous experimental validation loops, ensuring that computational predictions translate into actionable medicinal chemistry decisions.

90%+

Preclinical attrition
addressed by our pipeline

6×

Faster screening vs.
traditional wet-lab only

\$2.6B

Average cost to market we
help reduce

3

Core AI services for drug
discovery

Academic Grounding

Hyperionsoft's drug discovery practice is led by faculty affiliated with the USC Alfred E. Mann School of Pharmacy and Pharmaceutical Sciences, ensuring that every AI model is evaluated against pharmaceutical science first principles — not only computational benchmarks.



SECTION 02

Technical Architecture & Model Benchmarks

Architecture Overview

Our drug discovery platform is built on a modular, cloud-agnostic architecture deployable on AWS, Google Cloud, or Microsoft Azure. Each service layer is independently scalable and integrates via REST APIs or direct data pipeline connectors into existing laboratory informatics systems.

LAYER	COMPONENTS	TECHNOLOGY STACK
Data Ingestion	ChEMBL, PubChem, proprietary assay data, EHR-linked biobank feeds	Apache Spark, Delta Lake, AWS Glue / Azure Data Factory
Featurization	Molecular fingerprints (ECFP4/6), graph representations, 3D conformer ensembles	RDKit, DeepChem, OpenBabel, DGL-LifeSci
Modeling	Graph Neural Networks, Transformer-based LLMs, Bayesian optimization, diffusion models	PyTorch Geometric, HuggingFace, DiffSBDD, GuacaMol
Protein Structure	Target structure prediction, binding pocket analysis, molecular docking	AlphaFold 2/3 DB, AutoDock Vina, Schrodinger Glide (optional)
Validation	Prospective virtual screening, matched molecular pairs, scaffold activity cliffs	Custom validation harnesses, RDKit SAR analysis
Delivery	Interactive dashboards, ranked hit lists, structure-activity reports	Streamlit / Dash, PowerBI, REST API endpoints

Model Benchmark Summary

All models are evaluated on publicly available held-out test sets and, where available, cross-validated against proprietary client assay data. Benchmarks are updated semi-annually.

TASK	MODEL CLASS	BENCHMARK SET	METRIC	SCORE
Solubility (LogS)	GNN + Ensemble	ESOL (Delaney)	RMSE	0.41 log mol/L
hERG Cardiotoxicity	Transformer	ChEMBL hERG IC50	AUC-ROC	0.93
CYP3A4 Inhibition	MPNN	PubChem CYP3A4	AUC-ROC	0.91
BBB Penetration	GNN	B3DB Dataset	AUC-ROC	0.94
Oral Bioavailability	RF + GNN	Internal + Ma et al.	AUC-ROC	0.82
Repurposing Recall	Graph Attention	Gottlieb et al. DB	AUROC	0.86
De Novo Validity	Diffusion Model	ZINC250k hold-out	Validity	97.3%



De Novo Drug-likeness	Diffusion	ZINC250k hold-out	QED Score	0.74 avg
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Infrastructure Note

All GPU-accelerated training workloads are deployable on AWS p3/p4 instances, Google Cloud A100 nodes, or Azure NDv4 series. Inference endpoints support sub-500ms latency for single-molecule queries and batch processing of 100K+ compounds overnight. Data residency requirements (GxP, HIPAA, 21 CFR Part 11) are supported via private cloud or VPC-isolated deployments.

SECTION 03

Case Studies with Quantified Outcomes

The following engagements are presented in anonymized form to protect client confidentiality. Detailed references available upon execution of a mutual NDA.

CASE STUDY 01		ADMET-FIRST LEAD OPTIMIZATION		MID-SIZE ONCOLOGY BIOTECH	
Challenge	Client had 2,400 synthesized compounds in a kinase inhibitor program. Late-stage ADMET screening was eliminating ~70% of candidates, creating a costly cycle of synthesis and failure. Time from synthesis to ADMET decision averaged 11 weeks.				
Approach	Deployed multi-task GNN trained on client's proprietary assay history (8 ADMET endpoints simultaneously). Integrated into the compound registration system so every new structure received predicted ADMET flags before synthesis was authorized.				
Outcomes	• Synthesis-to-decision time reduced from 11 weeks to 4 days • Preclinical attrition due to ADMET failure reduced by 61% • 3 previously deprioritized scaffolds re-ranked to active development • Estimated 18 months saved before IND filing • R&D cost avoidance estimated at \$4.2M over program lifetime				
61%		4 days		\$4.2M	
Reduction in ADMET attrition		Synthesis-to- decision time		Estimated cost avoidance	
				18 mo	
				Time saved to IND	

CASE STUDY 02	DRUG REPURPOSING FOR RARE DISEASE	TOP-20 GLOBAL PHARMA
Challenge	Client held a library of 340 approved and Phase II-failed compounds with no active programs. The rare disease indication of interest had no approved therapies and a patient population too small to justify a de novo discovery investment.	
Approach	Built a heterogeneous biomedical knowledge graph linking compounds, targets, pathways, phenotypes, and disease ontologies using data from DrugBank, STRING, DisGeNET, and client proprietary profiling data. Applied graph attention networks to score compound-indication pairs, filtered by mechanistic plausibility using AutoDock Vina docking against AlphaFold-predicted target structures.	
Outcomes	7 high-confidence repurposing candidates identified from existing library - Top candidate entered IND-enabling studies within 9 months of project start - Timeline to first-in-human estimated 4 years vs. 10+ years for de novo approach - Patent filing initiated on novel indication for 2 additional candidates - Project ROI projected at 12:1 vs. full de novo program cost	



7

Repurposing candidates
identified

9 mo

To IND-enabling studies

4 yrs

Est. time to first-in-human

12:1

Projected ROI vs. de novo
program

SECTION 03 — CONT.

Case Studies (continued)

CASE STUDY 03		DE NOVO DESIGN FOR NOVEL TARGET CLASS		ACADEMIC SPINOUT / SERIES A BIOTECH	
Challenge	Early-stage biotech had validated a novel protein-protein interaction (PPI) target with strong genetic evidence but no known small-molecule binders in any public database. Traditional HTS was not feasible given budget constraints (~\$2M Series A).				
Approach	Used structure-based generative design with a pocket-conditioned 3D diffusion model (DiffSBDD variant), seeded from AlphaFold-predicted binding interface. Generated 850K candidate structures in silico over 72 hours. Applied multi-stage funnel: drug-likeness filter → ADMET pre-screen → docking score → synthetic accessibility scoring.				
Outcomes	• 850,000 de novo structures generated; 214 selected for synthesis • 31 confirmed binders in primary SPR assay (14.5% hit rate vs. 0.1–1% HTS baseline) • Best-in-class lead: Kd = 180 nM, drug-likeness score 0.81, no hERG liability • Full design-to-confirmed-binder cycle completed in 11 weeks • Enabled Series B fundraise of \$28M with lead compound as anchor asset				
14.5% Hit rate vs. 0.1–1% HTS baseline		11 wks Design to confirmed binder		180 nM Best lead Kd (binding affinity)	
				\$28M Series B enabled by lead compound	



SECTION 04

Data Requirements & Integration Checklist

Successful AI drug discovery deployments are gated as much by data readiness as by model sophistication. The following checklist is provided to help prospective partners assess their current state and identify gaps before project scoping. Items marked Required are necessary for baseline service delivery; items marked Recommended significantly improve model performance.

Compound & Assay Data

COMPOUND & ASSAY DATA — Required / Recommended / Optional	
✓	Registered compound structures in SMILES or SDF format (minimum 500 for fine-tuning; 5,000+ preferred)
✓	Biological assay results linked to compound IDs (IC50, Ki, EC50, or binary active/inactive)
✓	ADMET assay data: at minimum solubility, permeability (PAMPA or Caco-2), and one toxicity endpoint
■	Dose-response curves or raw assay values (not just binary labels) for key endpoints
■	Batch/plate information for assay QC and systematic bias correction
■	Counterscreen and selectivity panel data for target deconvolution
■	Historical synthesis records with yield and purity for synthetic accessibility modeling

Target & Structural Data

TARGET & STRUCTURAL DATA — Required / Recommended / Optional	
✓	Target gene name, UniProt ID, and primary disease indication
✓	Protein sequence in FASTA format (AlphaFold structure generated if no crystal structure available)
■	Known co-crystal structures or cryo-EM density maps in PDB format
■	Binding pocket coordinates or residue list if known from mutagenesis or literature
■	Off-target panel profile (selectivity data against related family members)
■	Allosteric site information for non-orthosteric design campaigns

Repurposing-Specific Data

DRUG REPURPOSING DATA — Required / Recommended / Optional	
✓	Inventory of existing approved or clinical-stage compounds (with mechanism of action annotations)
✓	Disease ontology mapping (MeSH, MONDO, or ICD-10 codes) for target indication



■	Transcriptomic or proteomic profiling data from disease vs. healthy tissue (if available)
■	Existing phenotypic screening data in disease-relevant cell lines
■	Clinical outcome data from EHR or real-world evidence cohorts (de-identified)

Systems Integration Requirements

SYSTEM	INTEGRATION METHOD	NOTES
ELN (e.g., Dotmatics, CDD Vault)	REST API or direct DB connector	Bidirectional preferred for real-time prediction
Compound Registration (e.g., Perkin Elmer)	SMILES/SDF export or API hook	Required for pre-synthesis ADMET screening
LIMS / Assay Management	CSV export or API	Needed for model retraining loops
Cloud Data Warehouse (Snowflake, BigQuery)	Native connector	Preferred for large-scale batch inference
Identity / SSO (Okta, Azure AD)	SAML 2.0 or OAuth 2.0	Required for GxP-compliant deployments
Visualization (Spotfire, TIBCO)	Plugin or REST API	Dashboard delivery option

SECTION 05

Engagement Models & Timeline Expectations

Hyperionsoft offers three engagement models designed to match the maturity stage and risk tolerance of each client. All engagements begin with a complimentary one-day Discovery Workshop to assess data readiness, define success metrics, and establish a shared technical roadmap.

SPRINT

Proof of Value

6–10 weeks \$45K – \$95K

Ideal for organizations evaluating AI drug discovery for the first time or validating a specific hypothesis (e.g., 'Can AI predict our hERG liabilities?'). Delivers a working prototype, benchmark report, and go/no-go recommendation with full data back.

Key Deliverables: Single endpoint ADMET model or repurposing screen · Retrospective validation on client historical data · Executive summary and technical findings report · Model code and documentation (client-owned IP)

DEPLOY

Production Integration

3–6 months \$180K – \$450K

Full-scale deployment of one or more AI services into the client's existing discovery infrastructure. Includes model training on proprietary data, system integrations, user training, and a 90-day hypercare period post-launch.

Key Deliverables: Multi-endpoint ADMET suite or full repurposing platform · ELN/LIMS integration and real-time prediction API · User training and change management support · Model retraining pipeline (quarterly or triggered by new data)

PARTNER

Embedded AI Program

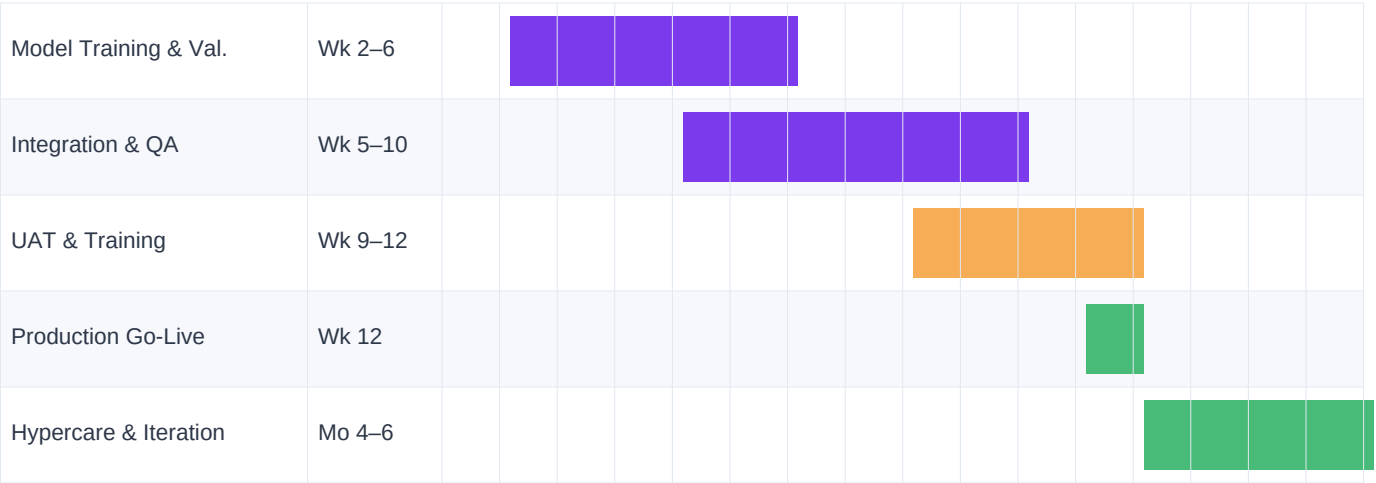
12–24 months Retainer + milestone

Long-term embedded partnership for organizations committing to AI-first drug discovery. Hyperionsoft functions as an extension of the internal team — attending program reviews, iterating models with each new assay cycle, and contributing to IND strategy.

Key Deliverables: All Sprint and Deploy deliverables · Dedicated AI scientist (FTE-equivalent) embedded in program · Roadmap prioritization and quarterly business reviews · Co-authorship on publications and patent filings where applicable

Indicative Project Timelines

PHASE	WINDOW	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	M4	M5	M6	M7
Discovery Workshop	Wk 1																
Data Assessment & Prep	Wk 1–3																



Timeline above reflects a typical Deploy engagement. Sprint engagements complete at Week 10; Partner engagements extend beyond Month 6.

SECTION 06

About Hyperionsoft & Academic Affiliation

USC Alfred E. Mann School of Pharmacy and Pharmaceutical Sciences

Hyperionsoft's drug discovery practice is led by faculty holding an appointment as Adjunct Professor at the USC Alfred E. Mann School of Pharmacy and Pharmaceutical Sciences, consistently ranked among the top 10 pharmacy schools in the United States. This affiliation ensures our AI methodologies are grounded in rigorous pharmaceutical science — encompassing medicinal chemistry, pharmacokinetics, toxicology, and clinical pharmacology. It also provides access to collaborative research infrastructure and a network of academic experts available for scientific advisory roles on client programs.

Our Technology Ecosystem

Hyperionsoft maintains active integrations and partnerships across cloud infrastructure and open-source scientific computing. Our stack is deliberately non-proprietary — clients own their models, their data, and their pipelines.

PARTNER / PROJECT	CATEGORY	ROLE IN OUR STACK
AWS (HealthLake, SageMaker)	Cloud Infrastructure	Primary compute, HIPAA/GxP-compliant data storage, model serving
Google Cloud (Healthcare & Life Sciences)	Cloud Infrastructure	AlphaFold integration, Vertex AI for model training, BigQuery for compound libraries
Microsoft Azure (Azure ML)	Cloud Infrastructure	Enterprise client deployments, Azure OpenAI for report generation
RDKit	Open Source — Cheminformatics	Core molecular featurization, fingerprinting, structure validation across all services
DeepChem	Open Source — Deep Learning	Model architectures for ADMET prediction and molecular property tasks
AlphaFold (DeepMind / EMBL-EBI)	Open Source — Structural Biology	Protein structure prediction for target-based design and docking campaigns

Next Steps

We recommend beginning with a no-cost, one-day Discovery Workshop — a structured session where our team reviews your current data assets, defines 2–3 high-priority AI use cases, and produces a written readiness assessment. Most clients proceed from Workshop to scoped proposal within two weeks.



**Request a Discovery
Workshop**
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ery**
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